

Jaime López Ruiz

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Portfolio: jaimelopezruiz.github.io | github.com/jaimelopezruiz | linkedin.com/in/jaimelopezruiz

EDUCATION

Imperial College London

Oct 2022 – Jun 2027

MEng Mechanical Engineering with Nuclear Engineering and a Year in Industry - Expected First Class Honours

- Relevant coursework: Mechatronics, Robot Dynamics & Control, Advanced Applied Optimisation, Machine Learning, Dynamics & Vibrations, Nuclear Fusion & Materials

Runnymede College, Madrid

2020 – 2022

- A-Levels: A*A*AA | Spanish Baccalaureate: 13.32/14 (95%)

EXPERIENCE

PerkinElmer Ltd.

Sep 2025 – Aug 2026

Mechanical Design Engineer (12-month Industrial Placement) | Global scientific instrumentation manufacturer, High Wycombe, UK

- Achieved 0.001° angular resolution in the optical path of a spectroscopy instrument by designing a coupled, synchronised precision motor assembly with a pre-loaded friction-drive mechanical reduction
- Programmed the assembly's motor control interface in Python over I2C and integrated it into the instrument, collaborating with optical scientists and electrical engineers
- Cut production test time by over 4 hours (30%) per instrument by designing and manufacturing custom test jigs and assembly accessories
- Owned components from concept to delivery: rapid prototyping (3D printing, machined parts) and procurement of standard, custom and motor components, applying GD&T, FMEA and poka-yoke

University of Oxford

Jun 2025 – Aug 2025

Undergraduate Researcher (UKAEA FOSTER Grant)

- Investigated irradiation-induced damage of SiC under nuclear reactor conditions using Raman spectroscopy; built a Python pipeline for spectral preprocessing and curve fitting
- First-author manuscript in preparation for peer-reviewed publication

Visual50

Jun 2024 – Sep 2024

Machine Learning Engineer Intern | Network optimisation software company, Madrid, Spain

- Developed a Python tool for IP network congestion detection from a network-optimisation product, using ML classification and reinforcement learning techniques

PROJECTS

Markerless Robot Arm Teleoperation

May 2026 - Jun 2026

Personal project

- Built a markerless teleoperation pipeline mapping monocular MediaPipe pose estimation to real-time joint commands for a 5-DOF SO-101 robot arm; no wearables, IMUs or fiducial markers
- Wrote a from-scratch Product-of-Exponentials kinematics library (custom URDF parser, FK, Jacobians, damped least-squares IK), numerically validated with condition-number analysis and convergence-rate benchmarking

Reusable Liquid Bipropellant Rocket

Oct 2024 – Jun 2025

Lead Feed System & Structures Engineer, Imperial College Rocketry Society

- First student rocket in UK history to hot fire vertically: led design, build and test of the propellant feed system and body structure for a 1 kN-thrust, 10 kN-s-impulse liquid bipropellant rocket
- Piston-driven tank held leak-proof at 80 bar test pressure for 60 minutes; two successful hot fires, the second meeting the 10 s sustained-thrust target at 78% of design mass flow

Autonomous Maze-Solving Buggy | Imperial College

2024 – 2025

- Bare-metal C on PIC18 (team of 2): I2C colour-sensor driver with calibrated gain/integration timing and fixed-point RGB normalisation classifying 9 instruction cards; stack-based path recording inverted to retrace the maze autonomously

LEADERSHIP & INVOLVEMENT

Vice-President, Spanish Society, Imperial College

2024 – 2025

- Represented an 800+ member Spanish community; organised a £2,500 fundraiser for Valencia flood disaster relief

Chassis Engineer (Brakes), Imperial College Formula Student

Oct 2023 – Jun 2024

- Designed and built the brake-oil reservoir bracket and headrest chassis mount, cutting component weight by 30%

CERTIFICATIONS

- Modern Robotics: Mechanics, Planning, and Control Specialization - Northwestern University (Coursera) - Ongoing
- Machine Learning with Python - MITx | Nuclear Energy: Science, Systems and Society - MITx

SKILLS

Technical: Python, C, MATLAB, SolidWorks, GD&T, FEA, FMEA / poka-yoke, I2C & embedded interfaces, Arduino

Languages: Spanish (native), English (native), French (fluent - DALF C1)